# DATA ENGINEERING

# ASSIGNMENT - 01

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Q1. Discuss the evolution of the data engineering over the past decade.

Ans:- Evolution of the data Engineer -

The evolution of data engineering reflects the changing ways organizations store, process, and derive value from data. Although technologies have changed over time, many core ideas continue to reappear.

1. Early Days (1980-2000) : Data Warehousing

The roots of data engineering lie in data warehousing which emerged in the 1980's.

→ with the development of relational databases and structured Query language (SQL) began storing & analyzing structured business data for reporting & Business intelligence (BI).

→ Data-modelling techniques are still widely used today

→ MPP (Massively Parallel Processing) databases enabled Scalable analytics.

→ Roles such as ETL developer, BI engineer, and data warehouse engineer emerged.

→ Data warehouse engineering became the foundation of modern data engineering.

2. Early 2000's : Birth of Contemporary Data Engineering

Due to data growth traditional databases could no longer scale efficiently which led to the need for new, scalable, cost-effective, and reliable data systems.

→ Key developments included :

- Widespread availability of cheap commodity hardware
- Rise of distributed storage & computation.
- Emergence of the big data era, characterized by Volume, velocity, and variety.

→ Technologies like GFS (Google File System), MapReduce, AWS, Hadoop are introduced.

3. 2000's - 2010's : Big Data Engineering Era

The hadoop ecosystem expanded rapidly with many tools. This period marked the rise of the big data engineer, who required strong software development skills and deep knowledge of distributed systems.

→ Shift from traditional enterprise tools to code-first engineering.

→ Maintenance of large on-premise data clusters.

→ Significant operational complexity and high maintenance costs.

→ Technologies like NoSQL, Spark, Kafka are introduced

4. 2020's : Engineering for the data lifecycle

In the current era, data engineering has evolved toward data lifecycle engineering. Engineers now rely on cloud-managed, modular, and highly abstracted tools rather than maintaining low-level infrastructure.

→ Modern data engineers focus on:

- DataOps and Orchestration.
- Data quality, governance, and security.

Q2. How does a data engineer's work differ when collaborating with data analysis versus data scientists?

Ans:- A data engineer works at the center of the data ecosystem and collaborates closely with both data analysts and data scientists. However, the nature of this collaboration differs based on the goals, tools, and outcomes expected from each role.

COLLABORATION WITH DATA ANALYST

Data analysts focus on understanding business performance, trends, and metrics using historical and current data. They typically work with structured data stored in data warehouses or data lakes and rely on SQL queries, spreadsheets, and BI tools.

When collaborating with data analysts, the data engineer's responsibility includes:

- Building and maintaining reliable data pipelines for analytical workloads.
- Ensuring data quality, consistency and correctness

- Creating data models that support reporting and dashboards

The primary goal of the data engineer in this collaboration is to provide well-structured, trustworthy data that enables accurate business reporting and decision-making.

COLLABORATION WITH DATA SCIENTISTS

Data scientists focus on forward-looking tasks, such as building predictive models and generating recommendations. Their work with large-scale, diverse, and frequently updated datasets

When collaborating with data scientists, a data engineer:

- Automates data collection, cleaning, and prepation processes.
- Builds scalable data infrastructure to handle large data volumes
- Reduces dependence on manual workflows and local development environments

The key objective of the data engineer in this collaboration is to help data scientists move models from experimentation to production efficiently & reliably.

Q3 What is a data maturity model? Discuss its levels and explain how organizations can progress through the model.

Ans: DATA MATURITY MODEL

A data maturity model describes the progression of an organization in effectively using data as a strategic asset. It reflects how well data is collected, managed, integrated, and utilized across the organization to generate business value. Data maturity does not depend on the age or size of a company but on how data is leveraged as a competitive advantage.

Different maturity models exist, but a simplified and practical data maturity model consists of three stages:

Level 1 : Starting with data

* Early stage of data adoption.
* Unclear goals and ad hoc data requests.
* Small data teams.
* Data engineers act as generalists.
* Focus on building basic data architecture and foundation.

Level 2 : Scaling with data

* Formal data practices established
* Scalable and robust data architectures.
* Adoption of DevOps and DataOps
* Data engineers specialize and support ML systems

Level 3 : Leading with data

* Organization is fully data-driven.
* Automated pipelines and self-service analytics.
* Strong data governance and quality practises.
* Data engineers build custom tools for competitive advantage.

Conclusion :

The data maturity model provides a structured path for organizations to evolve from basic data usage to data driven leadership, with the role of the data-driven ~~leadership~~ engineer becoming increasingly strategic at higher maturity level.